Self-Tuned Rejection Sampling within Gibbs and a Case Study in Small Area Estimation

Guide to Replication Materials

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1 Introduction

This document describes replication materials for the article "Self-Tuned Rejection Sampling within Gibbs and a Case Study in Small Area Estimation". It includes an R package, code for the two simulations, and two data analysis.

Section 2 presents an outline of the contents of the repo. Section 3 describes setup to run the replication materials. Section 4 gives an overview of the `saevws` package, including installation and basic usage. Section 5 lists the scripts used for the conditional simulation, posterior simulation, and data analysis. Section 6 gives a comprehensive list of packages and versions used in the computational studies, for reproducibility. Section 7 provides from details about the adaptive Metropolis-Hastings algorithm used for comparison in the data analyses.

2 Outline

Here is an outline of the contents of the `saevws-replication` repo.

- `guide`: contains the present document and the source files used to generate it.
- `joint`: materials for the "Joint SAE model" based on You (2021).
  - `01-sim-cond`: simulation generating from a single conditional of the SAE model under consideration.
  - `02-sim-post`: simulation generating from the posterior of the SAE model under consideration.
  - `03-sae`: data analysis with the SAIPE dataset.
  - `data`: the SAIPE dataset and codes / inputs to assemble it.
  - `shared`: defines R functions used in several seconds of
- `saevws`: an R package with the Gibbs samplers and supporting functions. See Section 4 of the present document.
- `unmatch`: materials for the "Unmatched SAE" model on You and Rao (2002).
  - `01-generated`: analysis based on the generated dataset.
  - `data`: a generated dataset based on a model in You and Rao (2002).

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3 Setup

A working installation of R (R Core Team 2026) is required with the following packages (and their dependencies).

- `coda` (Plummer et al. 2006)
- `jsonlite` (Ooms 2014)
- `knitr` (Xie 2014)
- `mcmcse` (Flegal et al. 2021)
- `Rcpp` (Eddelbuettel and Balamuta 2018),
- `statmod` (Smyth 2005)
- `tidyverse` (Wickham et al. 2019),
- `xtable` (Dahl et al. 2019)

The above packages are available on CRAN at the time of this writing. Additionally, development versions of the `fntl` (Raim 2024) and `vws` (Raim 2026) packages are needed. Here we demonstrate installing them using the `pak` package (Csárdi and Hester 2026).

```
R> pak::pkg_install("andrewraim/fntl@v0.1.3-saevws")
R> pak::pkg_install("andrewraim/vws@v0.3.1-saevws")
```

The `saevws` package included in the repo can now be installed. The following uses the `pak` package to accomplish this. (The placeholder `/path/to/` should be replaced with the actual path.)

```
R> pak::pkg_install("/path/to/saevws")
```

4 `saevws` R Package

The `saevws` package exports several functions for use in R. The primary functions are described here. See the associated manual page for each function for more information.

The following are for the Gibbs sampler in the Joint SAE example.

```
gibbs_joint(y, s2, X, Z, df, init = init_joint(m = length(y), d1 = ncol(X), d2 =
    ncol(Z)), control = control_joint(), fixed = fixed_joint())

control_joint(R = 1000, burn = 0, thin = 1, report = R + 1, save_latent =
    integer(0), record_mem = FALSE, inner = control_inner())

init_joint(m, d1, d2, beta = NULL, gamma = NULL, phi2 = NULL, tau2 = NULL,
    sigma2 = NULL, theta = NULL)

fixed_joint(beta = FALSE, gamma = FALSE, phi2 = FALSE, tau2 = FALSE, sigma2 =
    FALSE, theta = FALSE)
```

The following are for the Gibbs sampler in the Unmatched SAE example.

```
gibbs_unmatch(y, sigma, X, init = init_unmatch(m = length(y), d = ncol(X)),
    control = control_unmatch(), fixed = fixed_unmatch())

control_unmatch(R = 1000, burn = 0, thin = 1, report = R + 1, save_latent =
    integer(0), record_mem = FALSE, inner = control_inner())

init_unmatch(m, d, beta = NULL, tau2 = NULL, mu = NULL)

fixed_unmatch(beta = FALSE, tau2 = FALSE, mu = FALSE)
```

The following is used to specify the within-Gibbs sampling method in both examples.

```
control_inner(tol_suff = 0.01, tol_merge = exp(-100), max_rejects = 1e+06,
    method = c("vws-tune", "vws-basic", "imh", "arms", "amh"), N = 50, tune =
    1e+06, amh_varprop_init = 25, amh_varprop_c = 2.4, amh_varprop_eps = 0.05)
```

These functions are used for plotting.

```
traceplots(x, xlab = "")

plot_comps(x, burn, tol = 0.01)

plot_rejects(x, burn, tol = 0.2)

plot_tunes(x, burn, tol = 0.1)
```

5 Scripts to Reproduce Results in the Article

The following codes in `joint/01-sim-cond` produce the conditional simulation study in Section 6.1 of the article.

- `imh.R`: Simulations with IMH.
- `vws.R`: Simulations with VWS.

The following codes in `joint/02-sim-post` produce the posterior simulation study in Section 6.2 of the article.

- `gen.R`: Generate folder structure for the study. Each level of the study is placed into a separate folder.
- `sim.R`: Run one level of the study.
- `analyze.R`: Analyze the results from the completed study and produce summaries.

The `joint/03-sae` folder contains a single script `saipe.R` for the analysis in Section 7 of the article.

The `joint/data` folder contains the SAIPE dataset and scripts and inputs to construct it.

- `saipe.csv`: the prepared dataset.
- `cntysnap.csv`: Supplemental Nutrition Assistance Program (SNAP) data for 2022, to be used as a covariate with 2023 SAIPE data.
- `co-est2023-alldata.csv`: 2023 population estimates from the U.S. Census Bureau's Population Estimates Program. The data may be obtained from the Census Bureau website.
- `build-data.R`: Code to construct the dataset `saipe.csv`.

The `unmatch/01-generated` folder contains a single `unmatch.R` for the analysis in the supplement.

The `unmatch/data` folder contains the (simulated) dataset for the Unmatched SAE model, along with scripts and inputs to construct it.

- `build-data.R`: Code to construct the dataset `unmatch.csv`.
- `unmatch.csv`: the prepared dataset.

6 Session Information

Results in the paper were produced in an R environment with the following packages and versions.

```
library(coda)
library(jsonlite)
library(knitr)
```

```r
library(mcmcse)
library(Rcpp)
library(statmod)
library(tidyverse)
library(xtable)
library(vws)
```

```r
sessionInfo()
```

```
R version 4.6.1 (2026-06-24)
Platform: x86_64-pc-linux-gnu
Running under: Debian GNU/Linux forky/sid

Matrix products: default
BLAS:   /usr/lib/x86_64-linux-gnu/blas/libblas.so.3.12.1
LAPACK: /usr/lib/x86_64-linux-gnu/lapack/liblapack.so.3.12.1;  LAPACK version 3.12.0

locale:
 [1] LC_CTYPE=en_US.UTF-8       LC_NUMERIC=C
 [3] LC_TIME=en_US.UTF-8        LC_COLLATE=en_US.UTF-8
 [5] LC_MONETARY=en_US.UTF-8    LC_MESSAGES=en_US.UTF-8
 [7] LC_PAPER=en_US.UTF-8       LC_NAME=C
 [9] LC_ADDRESS=C               LC_TELEPHONE=C
[11] LC_MEASUREMENT=en_US.UTF-8 LC_IDENTIFICATION=C

time zone: America/New_York
tzcode source: system (glibc)

attached base packages:
[1] stats     graphics  grDevices utils     datasets  methods   base

other attached packages:
[1] saevws_0.2.0

loaded via a namespace (and not attached):
 [1] gtable_0.3.6       jsonlite_2.0.0     dplyr_1.2.1        compiler_4.6.1
 [5] brio_1.1.5         tidyselect_1.2.1   Rcpp_1.1.1-1.1     ellipse_0.5.0
 [9] stringr_1.6.0      scales_1.4.0       yaml_2.3.12        fastmap_1.2.0
[13] lattice_0.22-9     coda_0.19-4.1      ggplot2_4.0.3      R6_2.6.1
[17] generics_0.1.4     knitr_1.51         tibble_3.3.1       fftwtools_0.9-11
[21] pillar_1.11.1      RColorBrewer_1.1-3 rlang_1.2.0        testthat_3.3.2
[25] stringi_1.8.7      xfun_0.59          S7_0.2.2           otel_0.2.0
[29] mcmcse_1.5-1       cli_3.6.6          magrittr_2.0.5     digest_0.6.39
[33] grid_4.6.1         vws_0.3.1          fntl_0.1.3         lifecycle_1.0.5
[37] vctrs_0.7.3        evaluate_1.0.5     glue_1.8.1         farver_2.1.2
[41] rmarkdown_2.31     tools_4.6.1        pkgconfig_2.0.3    htmltools_0.5.9
```

7 Adaptive Metropolis-Hastings Algorithm

An Adaptive Metropolis-Hastings (AMH) method is implemented in our studies for comparison. The particular variant ("SCAM") is from Section 3.1 of Roberts and Rosenthal (2009) which is based on Haario et al. (2005). The objective is to draw $\boldsymbol{X} = (X_1, \ldots, X_k)$ from target π. A chain $\boldsymbol{x}^{(r)}$, $r = 0, 1, 2, \ldots$, produced by an MCMC method can approximate such draws. Using a Metropolis-within-Gibbs approach to draw component x_j, an "optimal" proposal distribution for Metropolis sampling would be $X_j^* \sim \mathrm{N}\big(x_j^{(r)}, (2.38)^2 \cdot \mathrm{Var}_\pi(X_j)\big)$. We can then accept $x_j^{(r+1)} = x_j^*$ with probability

$$\alpha(x_j^*, x_j^{(r)} \mid x_1, \dots x_{j-1}, x_{j+1}, \dots, x_k) = \min\left\{1, \frac{\pi(x_1, \dots, x_{j-1}, x_j^*, x_{j+1}, \dots, x_k)}{\pi(x_1, \dots, x_{j-1}, x_j^{(r)}, x_{j+1}, \dots, x_k)}\right\},$$

and let $x_j^{(r+1)} = x_j^{(r)}$ with the remaining probability. The constant 2.38 arises in MCMC theory. The variance $\mathrm{Var}_\pi(X_j)$ is computed under the target and therefore is generally not accessible except through approximation by an MCMC method. The objective of SCAM is to build up an approximation to $\mathrm{Var}_\pi(X_j)$ within the MCMC. Roberts and Rosenthal (2009) use SCAM with the proposal $\mathrm{N}(x_j^{(r)}, v_j^{(r)})$, where

$$v_j^{(r)} = \begin{cases} v_j^{(0)}, & r \le r_0, \\ s^2\{\tilde{\mathrm{V}}_j^{(r)} + \epsilon\}, & r > r_0, \end{cases}$$

with ϵ chosen to be a small positive number, $s = 2.4$ as a rounded version of the 2.38 optimality constant, initial value $v_j^{(0)}$, and empirical variance

$$\tilde{\mathrm{V}}_j^{(r)} = \frac{1}{r-1}\sum_{i=0}^{r-1}(x_j^{(i)} - \bar{x}_j^{(r)})^2,$$

with associated mean $\bar{x}_j^{(r)} = \frac{1}{r}\sum_{i=0}^{r-1} x_j^{(i)}$. Once the element $x_j^{(r)}$ of the chain is determined at the end of step r, the empirical mean and variance can be updated using the recursions so that it is not necessary to retain all of the draws:

$$\bar{x}_j^{(r+1)} = \begin{cases} x_j^{(0)}, & r = 0, \\ \frac{1}{r+1}\{r\bar{x}_j^{(r)} + x_j^{(r)}\}, & r > 0, \end{cases}$$

and

$$\tilde{\mathrm{V}}_j^{(r+1)} = \begin{cases} \{x_j^{(0)} - x_j^{(1)}\}^2/2, & r = 1, \\ \frac{r-1}{r}\tilde{\mathrm{V}}_j^{(r)} + \{\bar{x}_j^{(r)}\}^2 + \frac{1}{r}\{x_j^{(r+1)}\}^2 - \frac{r+1}{r}\{\bar{x}_j^{(r+1)}\}^2, & r > 1, \end{cases}$$

For the joint SAE application (You 2021), we use an exponential transformation to generate positive proposal values from the normal proposal distribution for σ_i^2. This results in the inclusion of a Jacobian term in the Metropolis ratio as follows. Let ϕ^* be a draw from proposal $\mathrm{N}(\phi^{(r)}, v_i^{(r)})$ with $\phi^{(r)} = \log \sigma_i^{2(r)}$. The density with respect to ϕ^* (with proportionality constant omitted) can be obtained by transformation as

$$f(\exp(\phi^*) \mid \text{—}) \propto f_{\mathrm{IG}}(\exp(\phi^*) \mid \kappa_i, \lambda_i) f_{\mathrm{LN}}(\exp(\phi^*) \mid \mu_i, \tau^2) \exp(\phi^*),$$

where the term $e^\phi = \frac{d}{d\phi}e^\phi$ is from the Jacobian of the transformation from σ_i^2 to ϕ. The acceptance ratio for Metropolis-Hastings is then

$$\varpi_i^{(r)} = \min\left\{1, \frac{f_{\mathrm{IG}}(\exp(\phi^*) \mid \kappa_i, \lambda_i) \cdot f_{\mathrm{LN}}(\exp(\phi^*) \mid \mu_i, \tau^2) \cdot \exp(\phi^*)}{f_{\mathrm{IG}}(\exp(\phi^{(r)}) \mid \kappa_i, \lambda_i) \cdot f_{\mathrm{LN}}(\exp(\phi^{(r)}) \mid \mu_i, \tau^2) \cdot \exp(\phi^{(r)})}\right\}. \tag{1}$$

For the unmatched SAE application (You and Rao 2002), we draw proposal variate $\mu_i^* \sim \mathrm{N}(\mu^{(r)}, v_i^{(r)})$ and accept with probability

$$\varpi_i^{(r)} = \min\left\{1, \frac{f_{\mathrm{LN}}(\mu_i^* \mid \boldsymbol{x}_i^\top \boldsymbol{\beta}, \tau^2) \cdot f_{\mathrm{N}}(\mu_i^* \mid y_i, \sigma_i^2)}{f_{\mathrm{LN}}(\mu^{(r)} \mid \boldsymbol{x}_i^\top \boldsymbol{\beta}, \tau^2) \cdot f_{\mathrm{N}}(\mu^{(r)} \mid y_i, \sigma_i^2)}\right\}. \tag{2}$$

Note that the Gaussian proposal cancels from both (1) and (2) because of symmetry.

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